

5 (**long**) Figure 5 shows the design of a reinforced concrete beam, with a square section of side L and four steel reinforcing bars of circular cross section, diameter D , placed symmetrically with their centres a distance $2D$ from the adjacent sides. The dimensions of the beam and the material properties are as shown in the figure. It may be assumed that in tension the concrete has no strength.

- (a) Show that when the beam is subjected to bending such that the top of the beam is in compression, the location of the neutral axis of the beam, indicated by αL in the figure, is given by $\alpha = 0.267$. [14]
- (b) Calculate the second moment of area for the beam. [8]
- (c) If the steel has a yield stress of 350 MPa calculate the maximum bending moment that the beam can sustain before the steel yields. [8]

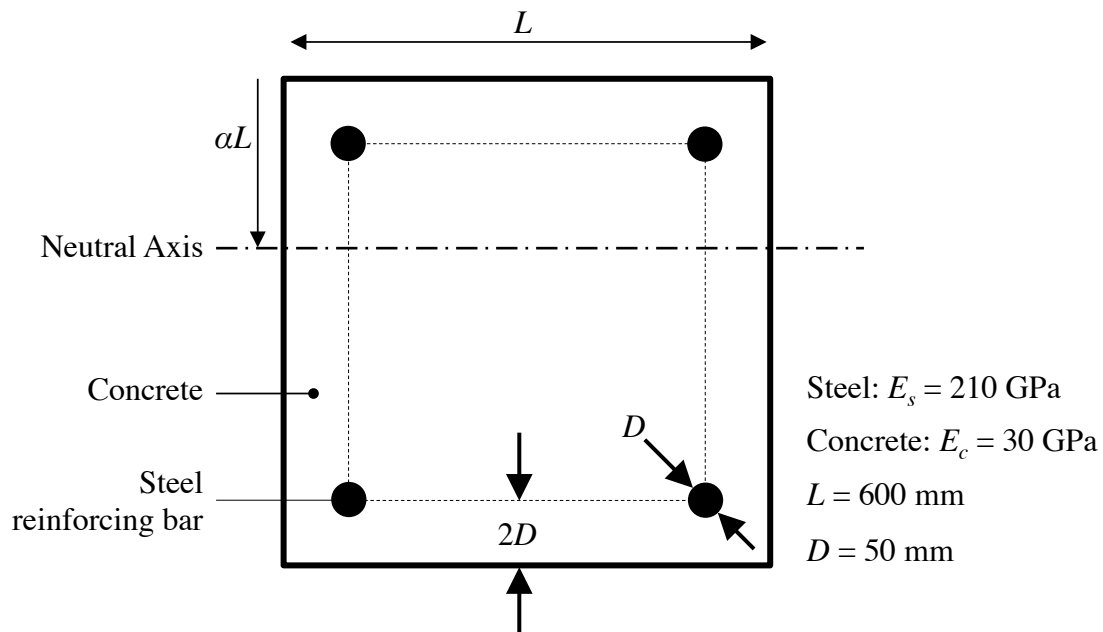


Fig. 5